

1. User needs & data needs:

Users	Social media analysts, social media users, Social Media Companies
User action (core value prop)	Retrieve and analyze relevant tweets based on user-defined queries
ML system output	Summarized insights based on retrieved tweets
ML system learning	Patterns of tweet relevance based on engagement metrics, semantic similarity, and user queries
Training dataset needed	Top 1000 tweets per query, User metadata, tweet engagement data
Key features needed in dataset	Username, location, Tweet engagement (likes, retweets, replies), Follower count, Date/time of tweet, Tweet type (original, reply, quote, retweet), Inferred sentiment (positive, neutral, negative)
Key labels needed in dataset	Relevance score based on engagement metrics, Semantic similarity score
Data formatting	user_id (string), tweet_id (string), username (lowercase), tweet_text (normalized text), engagement metrics
Real world data considerations	Does it include tweets from users with varying follower counts? The dataset includes tweets from users with varying follower counts, ensuring a diverse representation of social media influence. Does it capture tweets with media (images, videos, links) and their impact on engagement? Tweets with media (images, videos, links) are not explicitly stored in the dataset, as URLs are removed during preprocessing.
Data source key user questions	“How does the system know which tweets are relevant to my query?” The system determines tweet relevance by combining semantic similarity (via BERT embeddings) and engagement metrics (likes, retweets, replies). A ranking function in Pinecone ensures that retrieved tweets are contextually relevant to user queries.

Synthesize Your Core Data Needs:

Our product/service uses: Twitter dataset from Kaggle to provide social media users with relevant tweet insights.

Critical labels for our data include:

- Relevance score based on engagement metrics
- Semantic similarity score
- User-generated query context

We're aware of how the real world (e.g. time of year, changing trends) can impact the data used in our model.

To reflect the dynamism of the real world we made sure our data includes:

- Tweets from diverse user groups
- Real-world tweet data

2. Speak with a Domain Expert

Questions for Domain Experts:

- What data are important in your domain for social media analysis?
- What makes data usable vs. unusable in your domain?
- How are data collected in your domain (e.g., via APIs, web scraping, public datasets)?
- Do you have recommendations for data collection and/or labeling organizations?
- What problems occur with the data (e.g., missing values, duplicates, bias)?
- Are there any environmental and/or temporal circumstances that impact data collection (e.g., time-sensitive events, trending topics)?
- How easy or difficult is it to reuse data in your domain?
- What are the top 3-5 things people should be aware of when working with data in this domain?

3. Data Collection Weighted Matrix

	Weight	Existing dataset (no transformations)	Existing dataset (with transformations)	New dataset + Existing dataset	New dataset
Fit for use case Is this data appropriate for your users and use case?	3	0*3	4*3	1*3	0*3

Consider PII and Protected Characteristics: in some regions it's illegal to use them to make certain predictions. Are there any risks of the dataset excluding certain user groups? Have you used the Facets tool or some other tool/technique to evaluate the dataset for bias?					
Legality / Compliance What data standards are in place for compliance, licensing, documentation? See if you the dataset has a Data Card (or whether your team would need to create one)	3	1*3	4*3	0*3	0*3
Maintainability Does your team have a plan for maintaining the data post launch? How will data stay up to date over time?	2	0*2	4*2	1*2	0*2
Data collection effort How will the data be collected? How will your team ensure ethical data collection practices?	2	0*2	3*2	2*2	0*2
Cost What are the costs of choosing the most expedient data vs. the best data?	1	0*1	3*1	2*1	0*1
Total	11	3	41	11	0

4. Data Labelers + Task Design

Since we are using an existing dataset with transformations, our dataset is already labeled. Therefore, we do not require manual data labelers for this project. However, we must ensure that our data transformations maintain the integrity and relevance of the labels.

4.1 Who Are Our Labelers?

Although we do not employ manual labelers, our dataset relies on getting csv data from Kaggle. The labeling originates from:

- Twitter Metadata (retweets, likes, engagement metrics).
- Hashtag Monitoring (filtered hashtags related to the Ukraine-Russia conflict).
- We are downloading it from Kaggle which consists of multiple csv files containing data about relevant tweets.

4.2 Task Instructions & Label Quality Considerations

Since we are downloading and preprocessing this dataset locally, we must ensure:

- Consistency in label formats across different CSV files.
- Standardization of timestamps, hashtags, and engagement metrics.
- Deduplication of repeated tweets that might exist across multiple files.
- Handling of Missing or Corrupt Data (filling missing values or filtering outliers).

4.3 Task Design and Usability

Since we are using an already labeled dataset from Kaggle and performing local preprocessing, the traditional manual labeling process does not apply. However, we must ensure that our data transformation process maintains usability and consistency.

Performing the Task Ourselves

Before processing the dataset, we conducted self-testing by:

- Downloading and inspecting multiple CSV files to check for format consistency.
- Identifying missing values, duplicate tweets, and inconsistent timestamps.
- Ensuring engagement metrics (likes, retweets) are properly structured for later use.

Corrections Identified:

- Standardizing date and time formats.
- Handling encoding issues in text data.
- Removing duplicated tweets that exist across different CSV files.

*Regarding the table, it is not applicable to our project because we are working on a new dataset that requires labelling.

5. Data Disaster/Diligence Headlines

Since we are downloading multiple CSV files from Kaggle and preprocessing them locally, we need to identify potential data risks and implement safeguards to ensure the dataset is reliable, unbiased, and ethically sound.

1. Data Privacy

✗ Users Raise Concerns Over the Use of Public Twitter Data Without Consent.

✓ Our Preprocessing Ensures No Personally Identifiable Information (PII) is Stored or Processed.

Points Considered:

- ✓ The dataset comes from Kaggle and does not involve direct API scraping from Twitter.
- ✓ Only public tweets are included, ensuring compliance with data privacy policies.
- ✓ No personally identifiable information (PII) such as email addresses, phone numbers, or private messages are processed.

2. Data Exclusion

✗ Criticism Over Potential Bias Due to Filtering Only English Tweets.

✓ Filtering for English Tweets Ensures Consistency for NLP Processing, but We Consider Expanding to Multi-language Data.

Points Considered:

- ✓ Only English tweets are retained (`language == "en"`) to maintain consistency for NLP models.
- ✓ This approach removes multilingual tweets, which may exclude diverse perspectives.
- ✓ **Potential Future Expansion:** If required, preprocessing can be adjusted to include other languages for broader analysis.

3. Data Ethics

✗ Ethical Concerns Over the Use of Pre-labeled Data Without Validation.

✓ We Ensure Transparency by Properly Documenting Preprocessing Steps and Data Transformations.

Points Considered:

- ✓ Kaggle contributors provided the initial dataset → Bias detection measures are in place.
- ✓ We acknowledge and document the original data sources.
- ✓ No human labelers are used, but preprocessing includes checks to ensure data fairness and balance.

4. Data Transferability

✗ Risk of Using Social Media Data in a Context It Was Not Originally Collected For.

✓ Preprocessing Ensures Tweets Are Cleaned, Filtered, and Structured Specifically for Our NLP Tasks.

Points Considered:

✓ Kaggle data was originally collected for Ukraine-Russia conflict analysis, so we must ensure that it aligns with our ML goals.

✓ We remove special characters, URLs, and hashtags to make the text useful for NLP tasks.

✓ Data is structured in a way that makes it transferable to different Machine Learning models.

5. Data Fragility

✗ Pipeline Failure: Inconsistent CSV Formats and Missing Data Affect Processing.

✓ Preprocessing Handles Missing Values, Ensures Uniformity, and Logs Errors for Debugging.

Points Considered:

✓ Standardized processing scripts ensure uniform column names and formats across multiple CSV files.

✓ Missing values are either imputed or dropped based on their impact on the dataset.

✓ Logging mechanisms track errors and anomalies to prevent silent failures.